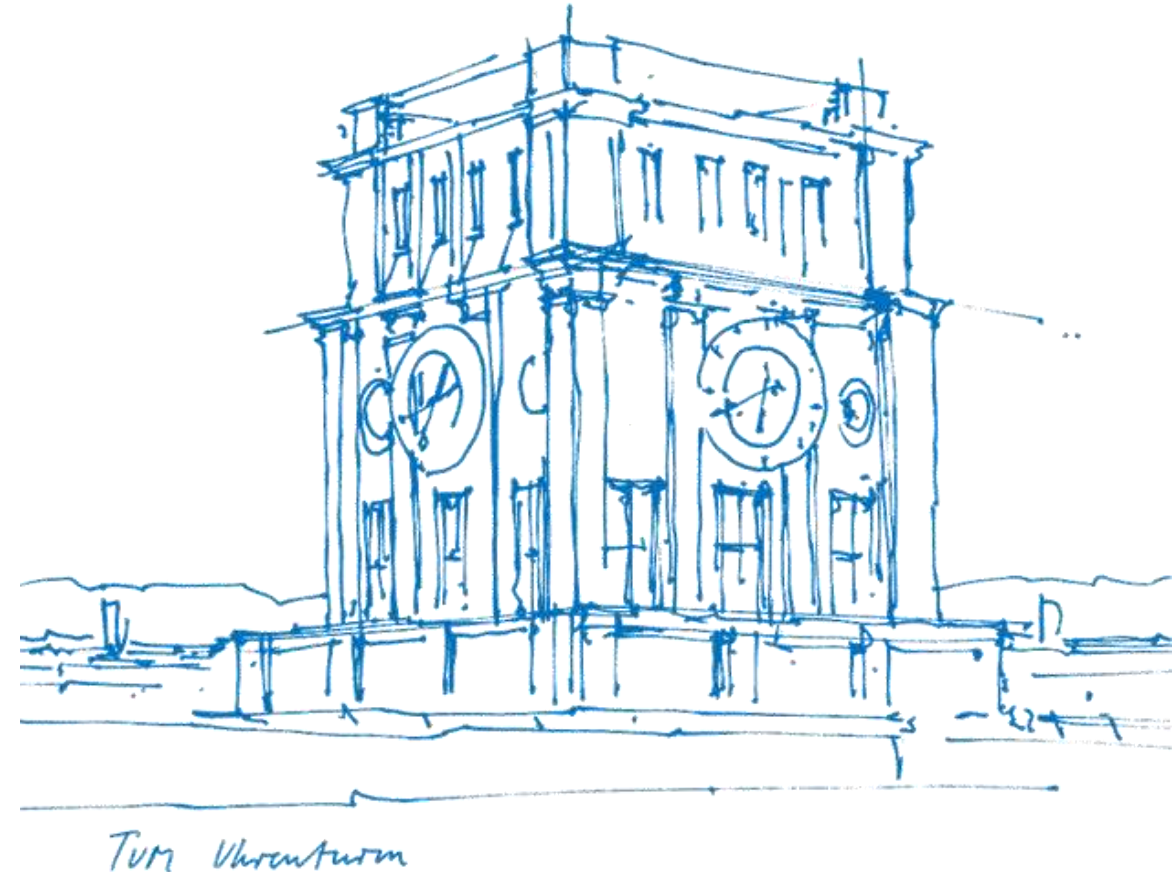


Tutorial 11

Hypothesis Testing

Xujie Zhao

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About Tutorial



■ **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
- 14:15-15:45 Tue, 00.08.053 (B42)

■ The slides are not guaranteed to be accurate! (Written by myself)

■ **Materials of this tutorial:** Please scan the QR-Code 

■ **Contact:** xujie.zhao@tum.de

- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

• Hypothesis Testing

- Null-Hypothesis (H_0) and Alternative-Hypothesis (H_1)
- Type I and Type II Errors
- One-sided and Two-sided Tests
- z-Test and t-Test
- Significance Level and Rejection Region
- Power of a Test
- Neyman-Pearson Lemma

II. Exercise Sheet

Exercise 1 (hypothesis test for a mean). A factory produces light bulbs, and the historical average lifetime is $\mu_0 = 1200$ hours with a known standard deviation of $\sigma = 150$ hours. A new manufacturing process is introduced, and the manager wants to test if the average lifetime has changed. A sample of $n = 36$ light bulbs produced by the new process has a sample mean lifetime of $\bar{x}_n = 1250$ hours. Assume the lifetimes are normally distributed.

- (i) State the null hypothesis H_0 and the alternative hypothesis H_1 for this test.
- (ii) Calculate the value of the appropriate test statistic.
- (iii) Would you reject the null at a significance level of $\alpha = 0.05$? What about at $\alpha = 0.001$? What does that mean for the confidence you should have in the rejection?

II. Exercise Sheet



Exercise 2 (Neyman-Pearson test for exponential distribution). Suppose we observe a single sample X from an exponential distribution $\text{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

- (i) Write down the likelihood ratio $\Lambda(x) = \frac{L(\lambda_1|x)}{L(\lambda_0|x)}$.
- (ii) According to the Neyman-Pearson lemma, the most powerful test of size α rejects H_0 if $\Lambda(x) > k'$ for some constant k' . Show that this rejection region is equivalent to $x < c$ or $x > c$ for some constant c . Determine which form is correct for $\lambda_1 > \lambda_0$.
- (iii) For a significance level α , determine the explicit form of the critical value c for the rejection region.
- (iv) What is the power of this test?

II. Exercise Sheet

Exercise 3 (one-sample t-test for a mean). A brewery claims that its bottles contain exactly 330 ml of beer. A skeptical student measures the contents of $n = 10$ randomly selected bottles and finds a sample mean of $\bar{x}_{10} = 328.5$ ml and a sample variance of $s_{10}^2 = 4 \text{ ml}^2$. Assuming the amount of beer in a bottle is normally distributed.

- (i) State the null and alternative hypotheses for testing if the true mean amount of beer is different from 330 ml.
- (ii) What is the appropriate test for this situation? State the test statistic and its distribution under the null hypothesis.
- (iii) Perform the test at a significance level of $\alpha = 0.05$. Should the student reject the brewery's claim?
[Hint: You may use that the 0.975-quantile of a t -distribution with 9 degrees of freedom is $t_{9,0.025} \approx 2.262$.]

II. Exercise Sheet

Exercise 4 (type II error and power of a z-test). Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

- (i) Determine the rejection region for this test. You may use that the 0.99-quantile of a standard normal distribution is $z_{0.01} \approx 2.33$.
- (ii) Calculate the probability of a type II error, β , if the true mean is actually $\mu_a = 11.5$.
- (iii) What is the power of the test against the alternative $\mu_a = 11.5$?

III. Additional Exercises

Exercise 5. A beverage manufacturer wants to determine whether their new Spezi recipe is better received by customers than their old recipe. According to previous surveys, the taste of the old recipe scored $\mu_0 = 7$ out of 10 taste points. A new survey is commissioned where 100 participants rate the taste of the new recipe on a scale from 1 to 10. You may assume that each person's rating is independent and identically distributed with unknown expected value μ . Using an appropriate hypothesis test, determine whether the claim $\mu \leq \mu_0$ can be rejected at a 1% significance level, given the following survey results:

Rating	1	2	3	4	5	6	7	8	9	10
Count	3	0	1	2	4	11	17	20	24	18

Note: You may approximate $t_{k,\alpha}$ for $k \geq 30$ with z_α , where z_α is the α -quantile of the standard normal distribution.

Hint: For observations $X_1, \dots, X_n$, sample variance is $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$, $n-1$ represents degree of freedom.

$$z_{0.99} \approx 2.31.$$

Define X_i as i -th rating from 100 participants, $E(X_i) = \mu$, $H_0: \mu \leq \mu_0$, use t-test since S is unknown.

$$\text{From the table, } \bar{X} = \frac{3 \times 1 + 2 \times 0 + \dots + 10 \times 18}{100} = 7.75, \quad S_{10} = \sqrt{\frac{3 \times (1-7.75)^2 + 0 \times (2-7.75)^2 + \dots + 18 \times (10-7.75)^2}{100-1}} \approx 1.99$$

$$T = \frac{\bar{X} - \mu_0}{S_n / \sqrt{n}} \approx \frac{7.75 - 7}{1.99 / \sqrt{100}} = 3.77, \text{ following the given hint, } t_{99, 0.99} \approx z_{0.99} \approx 2.31 < T, \text{ reject } H_0.$$

Materials

